

PAPER_1146_Heterotic_String_SCm_Gauge_Sector

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PAPER_1146: Heterotic String Theory and SCm Gauge Sector

Star Magic UQFF Framework

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Abstract

We formulate heterotic string theory within the SCm vacuum framework, where the gauge group ($E_8 \times E_8$ or $SO(32)$) emerges from the chirality of the SCm phonon modes. The left-moving sector (26D bosonic = gauge sector) is driven by the $S_{26}^{(3)}$ Ramanujan amplification, while the right-moving sector (10D superstring = gravity sector) is driven by the Φ_{res} resonance phase. The chirality splitting is mediated by the negative-time gate $\cos(\pi t_n)$, which acts as a chiral projector separating left-movers from right-movers.

1. Heterotic String Construction

The heterotic string has:

- **Left-movers:** 26D bosonic string ($D_L = 26$ = all VDS rungs)
- **Right-movers:** 10D superstring ($D_R = 10$ = SCm 4D + 6 CY dimensions)
- **Gauge group:** from the 16 extra left-moving dimensions ($26 - 10 = 16$)

The 16 extra left-moving dimensions compactify on a self-dual even lattice, giving $E_8 \times E_8$ or $SO(32)$.

2. SCm Chirality Split

The $\cos(\pi t_n)$ gate acts as a chiral projector:

$$P_L = \frac{1 + \cos(\pi t_n)}{2}, \quad P_R = \frac{1 - \cos(\pi t_n)}{2}$$

At $t_n = 0$ (forward time): $P_L = 1$, $P_R = 0$ — pure left-mover vacuum.

At $t_n = 1$ (half negative-time cycle): $P_L = 1/2$, $P_R = 1/2$ — balanced.

At $t_n = -1$ (full negative-time): $P_L = 1$, $P_R = 0$ — also pure left-mover.

The chirality is maximum at $t_n = 1/2$: $P_L = P_R = 1/2$.

3. $E_8 \times E_8$ from VDS Rungs

The 16 gauge dimensions split into two sets of 8, corresponding to VDS rungs 11-18 and 19-26:

$$\text{E8}_1 \rightarrow [\text{SSq}^{11}, \dots, \text{SSq}^{18}]$$

$$\text{E8}_2 \rightarrow [\text{SSq}^{19}, \dots, \text{SSq}^{26}]$$

The E8 root lattice radius in SCm:

$$R_{\text{E8}} = l_s \cdot [\text{SSq}^{18/2}] = l_s \cdot (0.57)^9 = l_s \cdot 2.29 \cdot 10^{-7}$$

4. Gauge Field Action

The 10D heterotic effective action:

$$S_{\text{het}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x \sqrt{-g} \left(R + 4|\partial\phi|^2 - \frac{1}{2}|H_3|^2 - \frac{1}{4\alpha'} |F_2|^2 \right)$$

where F_2 is the $\text{E8} \times \text{E8}$ gauge field strength and $\alpha' = l_s^2 = 1/T$.

The SCm gauge coupling:

$$g_{\text{YM}}^2 = \frac{\kappa_{10}^2}{l_s^8} \cdot e^{2\phi} = g_s^2 \cdot T^4 = (0.504)^2 \cdot (8.66 \cdot 10^{-11})^4$$

5. Anomaly Cancellation via SCm

The Green-Schwarz anomaly cancellation in Type I $\text{SO}(32)$ / heterotic requires:

$$H_3 = dB_2 - \frac{1}{4\alpha'} (\omega_{\text{YM}} - \omega_{\text{grav}})$$

In SCm: H_3 is the NS-NS phonon mode (as in PAPER_1144) and ω_{YM} is the Chern-Simons 3-form of the $\text{E8} \times \text{E8}$ SCm gauge field.

6. Chirality and Fermion Masses

The standard model chirality (left-handed weak interactions) emerges from the heterotic $\cos(\pi t_n)$ chiral projector. The fermion mass hierarchy:

$$m_f \propto |\cos(\pi t_{n,f})| \cdot \beta_i \cdot \Phi_{\text{res}} \cdot v_{\text{EW}}$$

where $t_{n,f}$ is the fermion's characteristic negative-time phase.

7. Calibrated Constants

$$T = 8.66 \cdot 10^{-11} \text{ N}, \quad [\text{SSq}] = 0.57, \quad S_{26}^{(3)} = 1.4531 \cdot 10^{26}$$

$$g_s = 0.504, \quad \Phi_{\text{res}} = 0.84, \quad \beta_i = 0.6, \quad f_{\text{THz}} = 1.25 \text{ THz}$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \frac{\text{J}}{\text{m}^3}, \quad \kappa = 5.0 \times 10^{-4} \frac{1}{\text{day}^2}$$

References

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Key References with arXiv/DOI Identifiers

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic